

# Manish Sagar Shetty

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## SUMMARY

Experienced Data Engineer with 3+ years of experience designing and optimizing scalable ETL pipelines and cloud data solutions on Azure and GCP. Proficient in SQL, PySpark, Databricks, and Snowflake, with a proven ability to enhance data pipeline performance and streamline workflows for large-scale analytics. Skilled in Power BI, Tableau, and ML frameworks to deliver actionable insights and support data-driven business strategies.

## TECHNICAL SKILLS

**Programming:** Python, PySpark, SQL (T-SQL, PostgreSQL, MySQL)

**Cloud Platforms:** Microsoft Azure (Data Factory, Databricks, Synapse, Data Lake), Google Cloud Platform (BigQuery, Dataflow), Microsoft Fabric, Snowflake

**Databases & Storage:** PostgreSQL, MySQL, Oracle AQL Server, SQL Server, BigQuery, MongoDB, Azure Data Lake, Delta Lake

**Machine Learning & AI:** Scikit-learn, TensorFlow, MLflow, MLOps, NumPy, Pandas, A/B Testing, Hypothesis testing, NLP, LLMs, Generative AI

**Data Visualization & BI:** Power BI, Tableau, Looker, Matplotlib, Seaborn, Plotly

**Other Tools/Frameworks:** Docker, Kubernetes, Git, Gitlab, Azure DevOps, CI/CD, Flask, DBT, Streamlit, Excel, Apache Airflow, Bitbucket

**Methodologies:** ETL/ELT Pipelines, Medallion Architecture, Data Quality Management, Agile Development

## WORK EXPERIENCE

### Data Engineer | Jul 2021 – Aug 2023

Cognizant | Hyderabad, India

- Implemented robust **SQL-based data processing workflows** with orchestrated pipelines, reducing operational costs by **15%** and improving decision-making accuracy by **22%**.
- Orchestrated **ETL pipelines** leveraging **PySpark** and **Databricks** following industry best practices, decreasing processing time by **40%** while maintaining **data integrity** and **quality standards**.
- Led migration of **on-premises data** infrastructure to **Azure Cloud** using **Medallion architecture** (Bronze, Silver, Gold layers), enabling **scalable** and **secure data storage** with improved accessibility.
- Developed comprehensive **data ingestion pipelines** using **Azure Data Factory** to automate **ETL processes** from diverse data sources into **Azure Data Lake**, with integrated **CI/CD deployment** through **Azure DevOps**.
- Optimized **50+ database queries** and data platform performance through strategic **query restructuring**, **indexing strategies**, and **Azure Databricks transformations**, achieving **35%** reduction in **query execution time**.
- Built **real-time data monitoring** systems and **Power BI dashboards**, transforming **raw data into actionable insights** and providing **stakeholder demo environments** for **data-driven** decision making.
- Established **monitoring** and **maintenance** policies using **Azure Data Factory** monitoring tools to identify pipeline failures, optimize performance by **15-30%**, and maintain **99.5% SLA compliance**.
- Mentored **6 junior team members** on **database optimization** and **data pipeline** best practices, reducing **data quality issues by 40%** and **onboarding time by 50%**, fostering continuous improvement culture.

### Data Analyst | Feb 2021 – Jul 2021

Cognizant | Hyderabad, India

- Executed **complex SQL queries** on **multi-terabyte databases** to extract **actionable customer insights**, contributing to a **12% increase in customer retention** and driving **targeted marketing strategies**.
- Created **comprehensive data visualization dashboards** via **Power BI** that transformed **raw data into intuitive visual stories**, increasing **stakeholder engagement** by **30%** and enabling **data-driven decision making** across departments.
- Collaborated with **business analysts** and **cross-functional teams** to define **KPIs** and **metrics**, establishing **standardized reporting methods** that increased **data consistency** by **25%**.
- Reduced error rates by **18%** and enhanced overall **data integrity** by implementing strong **validation methods** and performing **root cause analysis** on data quality concerns.
- Implemented **reporting solutions** using **SQL stored procedures** and **Python scripts**, reducing **manual reporting time** by **45%** and ensuring **timely delivery** of critical **business metrics**.
- Established **data governance protocols** and **documentation standards** for **25+ analytical processes**, ensuring **100% reproducibility** and reducing **knowledge transfer time by 65%** across **analytics team**.
- Performed **advanced statistical analysis** and **A/B testing** on **customer behavior** and **marketing campaigns**, providing **data-driven recommendations** that influenced **strategic decisions** and **product development**.

## Machine Learning Intern | Aug 2020 – Feb 2021

Ineuron | Hyderabad, India

- Developed **end-to-end machine learning models** using **scikit-learn** and **TensorFlow** that achieved **91% prediction accuracy** for **customer behavior analysis**, driving **data-driven insights** for **business strategy**.
- Performed **comprehensive data cleaning**, **feature engineering**, and **model deployment**, integrating **big data technologies** to enhance **data processing efficiency** and improve **model performance** across the **ML pipeline**.
- Deployed **REST API interfaces** through **Flask** and **Streamlit** to integrate **machine learning models** with **production systems**, enabling **real-time predictive capabilities** and **seamless user interaction**.
- Conducted **A/B testing** and **model validation experiments** to compare **algorithm performance** across multiple datasets, implementing **cross-validation techniques** that improved **model reliability** by **15%**.
- Collaborated with **data science team** to optimize **hyperparameters** and **feature selection** using techniques like **Grid Search** and **Random Forest importance**, resulting in **8% improvement in model accuracy** and reduced **training time**.
- **Containerized machine learning models** and **data processing pipelines** using **Docker**, improving the **scalability** and **portability** of **ML infrastructure** while ensuring **consistent model performance** across **development** and **production environments**.

## PROJECTS

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**Olympic Fabric Data Engineering** | [Project link](#) | **Tech** - Python, Pyspark, SQL, MS Fabric, Power BI

- Built comprehensive **data pipeline** processing **5+ Olympic datasets** (athletes, events, medals, countries) using **Fabric Lakehouse architecture** and **streamlined ETL workflows**.
- Delivered **actionable insights** for Olympic events through **ETL processes** and interactive **Power BI dashboards**.

**NYC Taxi Data Analytics** | [Project link](#) | **Tech** - Pyspark, SQL, Python, GCP, BigQuery, Looker

- Created end-to-end data pipeline ingesting raw trip data into **GCP Cloud storage** with Python Mage workflows and **automated ETL** on Compute Engine.
- Designed **interactive dashboards** visualizing **key performance metrics**, revealing critical **trip patterns** and **usage trends** for **data-driven decision making**.

**Nike Sales Analytics with Snowflake and Streamlit** | [Project link](#) | **Tech** - Python, SQL, Snowflake, Streamlit, Pandas, Plotly

- Designed **sales analytics dashboard** processing **retail dataset** with **customer segmentation** and **business intelligence visualizations** using Snowflake cloud platform.
- Produced **RFM analysis** and **market basket analysis** features showcasing **advanced analytics capabilities** and created **data export** functionality for retail operations.

**Netflix movie recommendation system** | [Project link](#) | **Tech** - Python, Pandas, Matplotlib, Seaborn, Scikit-learn, Flask

- **Built recommendation engine** achieving **85% accuracy** using cosine similarity on **9K+ movie dataset** with TF-IDF vectorization processing **50+ features**.
- Deployed **flask web application** serving **real-time recommendations** with **<2 second response time** and **interactive UI** supporting **concurrent user sessions**.

## EDUCATION

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**Master of Science in Data Science and Analytics** | Aug 2023 - May 2025

Florida Atlantic University | Boca Raton, Florida

## CERTIFICATIONS

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- Microsoft certified Azure AI fundamentals (AI-900) - [View Credential](#)
- Microsoft certified Azure Data fundamentals (DP-900) - [View Credential](#)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600) - [View Credential](#)
- SnowPro Associate: Platform Certification - [View Credential](#)
- Level - 4: Excel Black Belt (McGraw Hill) - [View Credential](#)
- SQL for Data Science (Coursera) - [View Credential](#)